

Aisha Bint Abi Baker School

Class 12 Elite: Physics

Chapter 10, Section 5: RLC Circuits Lab Report

Student Name:



Date:

Lab Objectives

1. Identify the frequency at which the circuit reaches maximum current (Resonance).
2. Measure individual component voltages (V_R, V_L, V_C) using probes.
3. Calculate the total impedance Z and verify source voltage using phasors.

1. Theoretical Background

The resonant frequency f_0 is where inductive and capacitive reactances are equal ($X_L = X_C$).

$$f_0 = \frac{1}{2\pi\sqrt{LC}}$$

At this point, the circuit is purely resistive, and current is at its maximum.

2. Data Collection (Fillable Table)

Use the virtual probes to measure the following for each frequency:

Freq (Hz)	Current (A)	V_R (V)	V_L (V)	V_C (V)
20 Hz				
40 Hz				
Resonance				
100 Hz				
150 Hz				

3. Analysis Questions

1. Calculate the theoretical resonant frequency (f_0) based on L and C :

2. Describe the behavior of V_L and V_C as you pass through resonance:

3. Verify if $V_{source} = \sqrt{V_R^2 + (V_L - V_C)^2}$ using your values at resonance:

4. Conclusion

Summary of findings: